

Market Structure and Cost Pass-Through in Retail Gasoline

Joshua Leinwand '27
ECON 162 Term Paper

Abstract

In this paper I seek to understand whether the structure of a local gasoline market changes the degree to which wholesale costs are passed through into retail prices. To do so, I use 604,817 station-week observations covering 6,556 stations, first estimating station and calendar-week fixed-effects models before interacting wholesale costs with measures of local concentration. I find that above-median HHI is associated with a 0.0070 increase in pass-through, while a one-standard-deviation increase in nearby rivals is associated with a 0.0026 decrease; both estimates are significant at the 1 percent level. The independent-station share, by contrast, has no statistically significant relationship with pass-through. The dynamic results suggest that concentration changes the timing of adjustment rather than simply increasing it at every point, while a complementary margin regression finds that a 0.10 increase in HHI is associated with roughly 0.8 cents per gallon in higher margins. Thus, although these estimates are conditional relationships rather than causal merger effects, local competitive structure nevertheless appears within the pricing conduct that consumers face.

1. Introduction

How much can the structure of a local market change the price that a consumer sees on a sign by the road? Gasoline may at first seem to be a fairly straightforward commodity, yet anyone moving between a rural area and a more densely populated one can tell that the market does not feel uniform from the ground. A nearby station, an independent seller, or simply another pump across the street may limit the freedom that a station has in choosing its price. In this paper I will seek to give this intuition an empirical form by asking whether local concentration changes the transmission of wholesale cost shocks in a way consistent with market power. Retail gasoline is useful for this question because competition is local and because the data contain repeated observations of both wholesale costs and retail prices. In this way, market definition, concentration, margins, and competitive constraints can be considered without first requiring a full structural model of demand.

Namely, the main hypothesis under consideration is that local market structure affects the pricing rule followed by gasoline stations. If a station in a concentrated market faces fewer close substitutes, then it may have more freedom to alter its markup as wholesale costs change. In a denser market, however, nearby rivals may discipline that same decision. Pass-through can therefore differ either in its timing or in its magnitude depending upon the competitive conditions around the station. This is not to say that every concentrated market is collusive, nor that every difference in pass-through constitutes an antitrust violation. Rather, the more limited question is whether local competitive conditions appear systematically within the pricing behavior of the stations themselves.

To consider this question, I use 604,817 station-week observations covering 6,556 stations. I first estimate the average relationship between wholesale costs and retail prices with station and calendar-week fixed effects. The station fixed effects remove the characteristics that make one station persistently different

from another, while the week fixed effects remove shocks shared across the gasoline market at a given time. I then interact wholesale costs with a high-HHI indicator, standardized HHI, nearby rival counts, and independent-station share. Finally, as another way to approach the question, I estimate whether average retail margins are higher in more concentrated local markets.

In this way, the exercise is closest in spirit to empirical merger analyses such as Staples, even though it neither evaluates a particular transaction nor predicts a post-merger price. The same antitrust tools are instead used to ask a prior question: does the structure of the market appear in the consumer-facing conduct of the firms that operate within it?

2. Antitrust Framework and Motivation

The main way to understand what this evidence can mean for antitrust is to consider the three questions underlying normative antitrust analysis. First, what is the theory of harm? Second, do the data show that the concern exists in a meaningful form? And finally, how would this evidence be used by an agency or court in practice? In the present setting, the concern is that local concentration may soften gasoline competition, allowing stations either to maintain higher margins or to adjust prices differently when their costs move. I therefore test whether this pattern appears in the data before considering what it can, and cannot, tell us about market power.

At the most basic level, Viscusi, Harrington, and Sappington define market power as the ability to profitably charge above the competitive level, identifying price-cost margins as one possible measure of that ability. From this definition, we can tell why margins are relevant to the paper. A high margin does not by itself prove unlawful conduct, but it does raise the question of whether firms face effective competitive constraints. The authors also warn us that a relationship between concentration and performance may be produced by efficiency, location, brand value, or unobserved demand. Thus, even a positive relationship between HHI and margins cannot be treated as market power without qualification.

The principal measure of concentration used here is the Herfindahl-Hirschman Index, or HHI, which is central to federal merger analysis. Although this paper does not evaluate a merger, the underlying question remains similar: does the structure of a local market help predict the exercise of market power? Kaplow and Shapiro direct our attention toward competitive effects rather than concentration for its own sake, while Williamson reminds us that concentration may also coincide with lower costs or other efficiencies. HHI must

therefore be understood as a starting point for the empirical inquiry, rather than as the conclusion that the inquiry is meant to establish.

3. Background and Literature

The antitrust literature first provides the general framework through which the empirical results can be understood. Viscusi, Harrington, and Sappington emphasize substitution in market definition, profitable pricing above competitive levels, and both the usefulness and the limits of HHI. The 2010 and 2023 Merger Guidelines similarly use geographic markets and concentration to organize the analysis of competitive effects. Yet, as Kaplow and Shapiro (2007) make clear, these measures are not meant to be applied mechanically. Their purpose is to help us understand competition, not to replace an analysis of it.

Now that the general antitrust concepts have been laid out, we are free to consider how they have been tested empirically. Baker's account of *FTC v. Staples* explains how price evidence was used to assess the competition between the merging firms, rather than merely to state that the market was concentrated. Ashenfelter, Hosken, and Weinberg's study of beer and Dafny, Duggan, and Ramanarayanan's study of health insurance similarly estimate the price effects associated with consolidation in actual markets. In each case, the structure of the market matters because it is connected to observed conduct.

To consider this same relationship in gasoline, both the outcome and the geographic market must be motivated. Borenstein, Cameron, and Gilbert (1997) show that retail gasoline prices respond asymmetrically to movements in crude oil, while Borenstein and Shepard (1996) connect retail margins to dynamic oligopoly pricing. Hastings (2004) demonstrates that independent retailers and vertical relationships can affect gasoline prices, suggesting that the identity of a nearby rival may matter alongside its simple existence. Houde (2012), by modeling commuting paths and spatial differentiation, further shows why gasoline competition must be understood as local. I adopt this premise, even though the available data do not allow me to estimate a structural model of spatial demand.

Finally, the broader pass-through literature explains why this particular relationship is worth studying. Weyl and Fabinger (2013) connect pass-through to incidence and imperfect competition, while Nakamura and Zerom (2010) show that incomplete pass-through may reflect markup adjustment, local costs, or pricing frictions. In this way, pass-through is more than a mechanical relationship between two prices. If it varies

across local gasoline markets, the variation may tell us something about firm conduct, demand, or the competitive constraints that stations face.

4. Data and Market Definition

The empirical work begins by combining two datasets. The first is a station-day panel containing retail gasoline prices, wholesale costs, taxes, freight costs, margins, station identity, and location. The second contains station characteristics, including brand, independent or major-chain status, amenities, and demographic variables measured at either the county or tract level. Taken together, the two datasets allow the pricing behavior of a station to be connected to both its own characteristics and the market that surrounds it.

Before moving into the regressions, I aggregate the station-day observations to the station-week level. This preserves the panel structure while reducing the noise created by daily reporting gaps and making the analysis more tractable. More importantly, the weekly level better matches the object under consideration. The question is whether retail prices adjust to recent wholesale costs, rather than whether small day-to-day reporting fluctuations create measured price movements.

The question of what constitutes the market around a station is more difficult. I construct local markets with latitude and longitude, using a three-mile radius in the preferred specification and one- and five-mile radii as robustness checks. This is an empirical approximation of the geographic-market problem discussed in the Merger Guidelines and by Viscusi, Harrington, and Sappington. In an ideal investigation, we would ask whether enough consumers would leave a proposed market after a small but significant and nontransitory price increase. Yet because these data contain neither quantities nor consumer switching, geographic proximity must serve as a practical substitute. The three radii are therefore not definitive boundaries, but competing approximations of the stations likely to constrain a focal station.

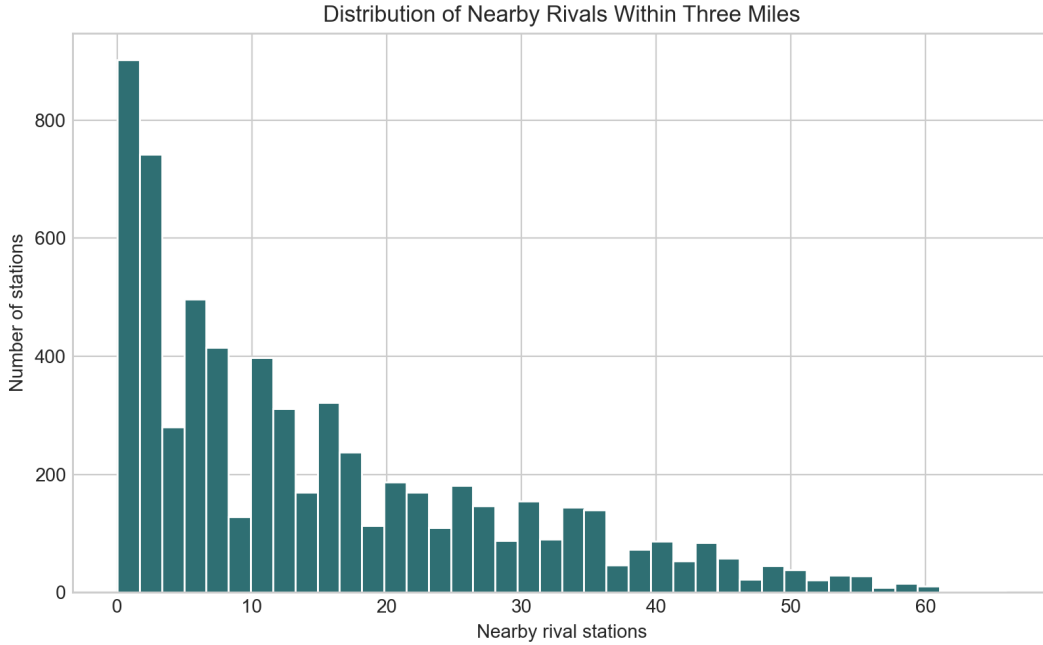
Using these geographic markets, I construct three principal measures: the number of nearby rival stations, local brand HHI, and the share of nearby rivals that are independent stations. The focal station is excluded from the rival count because it cannot be its own competitor. It is included in HHI, along with the nearby stations, because HHI describes the brand concentration of the full set of stations available around that location. These distinctions are small mechanically, but necessary for understanding what each measure actually represents.

Using the above definitions, Table 1 reports the summary statistics and Figure 1 shows the distribution of nearby rivals within the preferred three-mile radius. The central descriptive fact is immediately visible: gasoline stations do not face the same local competitive environment. Some are surrounded by many alternatives, while others operate with relatively few nearby rivals.

Table 1. Summary Statistics

Variable	Mean	Std. dev.	10thPercentile	50thPercentile	90thPercentile	N
Retail price	2.886	0.531	2.159	3.029	3.517	604,817
Wholesale cost	2.274	0.527	1.545	2.472	2.931	604,817
Retail margin	0.206	0.132	0.057	0.196	0.365	604,817
Local HHI, 3 miles	0.277	0.232	0.112	0.185	0.500	604,770
Nearby rivals, 3 miles	15.750	14.044	1.000	12.000	37.000	604,770
Nearby independent share, 3 miles	0.432	0.257	0.159	0.385	0.833	570,430
Major brand indicator	0.241	0.428	0.000	0.000	1.000	604,146
Independent station indicator	0.447	0.497	0.000	0.000	1.000	604,146
Tract income per capita	26619.625	11574.221	16313.000	23913.000	40521.000	602,738
Tract unemployment rate	8.668	4.510	3.700	8.000	14.500	602,632

Notes: The unit of observation is a station-week. Prices, costs, and margins are measured in dollars per gallon. Local market measures use stations within three miles of the focal station.



5. Empirical Strategy

The first regression considers the baseline relationship between wholesale costs and retail prices, and takes the following form:

$$P_{it} = \beta W_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

Here, P_{it} is the average retail price of station i in calendar week t , while W_{it} is the average wholesale cost faced by that station in the same week. The station fixed effect α_i removes persistent differences in location, brand, amenities, highway access, and customer base. The calendar-week fixed effect λ_t removes shocks shared by stations in a given week, and ε_{it} represents the variation that remains unexplained. Thus β measures within-station pass-through after common weekly movements have been removed. Because the observations for one station are likely to be related over time, standard errors are clustered by station.

Now that the baseline relationship has been established, local market structure can be introduced through the following interaction:

$$P_{it} = \beta W_{it} + \gamma(W_{it} \times C_i) + \alpha_i + \lambda_t + \varepsilon_{it}$$

The term C_i takes four different forms, each estimated in a separate regression: a high-HHI indicator, standardized HHI, standardized rival count, and standardized independent-station share. The high-HHI

indicator equals one when a station's local HHI is at or above the sample median for that radius. The other measures are standardized to have a mean of zero and a standard deviation of one. In the high-HHI model, β is the pass-through of below-median-HHI stations and γ is the additional pass-through of high-HHI stations. In the standardized models, β is pass-through at the sample mean of C_i , while γ is the change associated with a one-standard-deviation increase in that measure. A separate C_i term does not appear because market structure is fixed for a station and is therefore absorbed by α_i .

To consider whether the relationship occurs immediately or over time, I next estimate the dynamic specification:

$$P_{it} = \beta_0 W_{it} + \beta_1 W_{i,t-1} + \beta_2 W_{i,t-2} \\ + \gamma_0 (W_{it} \times H_i) + \gamma_1 (W_{i,t-1} \times H_i) + \gamma_2 (W_{i,t-2} \times H_i) + \alpha_i + \lambda_t + \varepsilon_{it}$$

In this equation, $W_{i,t-1}$ and $W_{i,t-2}$ are wholesale costs from one and two weeks earlier, and H_i is the three-mile high-HHI indicator. The β coefficients give the current and lagged pass-through of below-median-HHI stations. The γ coefficients give the additional current and lagged relationships of high-HHI stations. In this way, the separate coefficients tell us about timing, $\beta_0 + \beta_1 + \beta_2$ gives cumulative pass-through for the reference group, and $\gamma_0 + \gamma_1 + \gamma_2$ gives the cumulative difference for high-HHI stations.

Finally, to approach market power through retail margins rather than pass-through alone, I estimate:

$$\text{Margin}_{it} = \delta_1 \text{HHI}_i + \delta_2 \text{Rivals}_i + \delta_3 \text{IndependentShare}_i \\ + \theta_1 \text{Major}_i + \theta_2 \text{Independent}_i + \theta_3 (\text{Income}_i / 10,000) + \theta_4 \text{Unemployment}_i + \lambda_t + \varepsilon_{it}$$

All three market-structure measures in this equation are defined within three miles and enter at the same time. Major_i and Independent_i identify station type, Income_i is tract income per capita measured in units of \$10,000, and Unemployment_i is the tract unemployment rate. The model includes calendar-week fixed effects λ_t and standard errors clustered by station. Station fixed effects cannot be used here because they would absorb HHI, rival count, independent share, and the other characteristics that do not change over time. For this reason, the margin regression is best understood as complementary descriptive evidence, rather than as a stand-alone causal estimate.

It is also important to clarify what these regressions are not. Without quantities or an observed merger, I cannot estimate demand elasticities, diversion ratios, or a post-merger counterfactual. The exercise is therefore not a merger simulation. What it can do is test whether local concentration, rival density, and margins

are systematically related to pricing conduct. This is a smaller claim, but one that remains relevant to antitrust analysis.

6. Main Results

We are now free to consider the results, beginning with the baseline estimate in Table 2. The wholesale-cost coefficient is 0.5196 with a standard error of 0.0046, and is statistically significant at the 1 percent level. In practical terms, a one-dollar increase in a station's wholesale cost is associated with an increase of approximately 52 cents in its retail price once station and week fixed effects have been removed. Wholesale costs therefore move strongly with retail prices, even though the relationship remains below one-for-one pass-through.

Table 2. Baseline Wholesale-to-Retail Pass-Through

Variable	Baseline
Wholesale cost	0.5196*** (0.0046)
N	604,817
Within R2	0.158
Outcome mean	2.886

Notes: Clustered standard errors by station are in parentheses. The dependent variable is retail price. The model includes station fixed effects and calendar-week fixed effects.

Table 3 moves from the baseline relationship into the four three-mile interaction models, each estimated separately. In the high-HHI model, the wholesale-cost coefficient for below-median-HHI stations is 0.5157 and the high-HHI interaction is 0.0070; both are significant at the 1 percent level. Thus, the estimated pass-through of high-HHI stations is approximately 0.5227. In the standardized-HHI model, pass-through at mean HHI is 0.5193, while a one-standard-deviation increase in HHI changes it by 0.0021; again, both coefficients are significant at the 1 percent level. The standardized rival-count model gives a wholesale-cost coefficient of 0.5193 and an interaction of -0.0026, both significant at the 1 percent level. Finally, the independent-share model has a wholesale-cost coefficient of 0.5264, significant at the 1 percent level, but an interaction of only 0.0002 that is not statistically significant at conventional levels. In this way, HHI and rival count leave a modest trace in pass-through, while independent share does not provide the same evidence.

Table 3. Market Structure and Pass-Through, Main 3-Mile Market

Variable	High HHI	HHI, std.	Rival count, std.	Independent share, std.
Wholesale cost	0.5157*** (0.0048)	0.5193*** (0.0046)	0.5193*** (0.0046)	0.5264*** (0.0047)
Wholesale cost x high-HHI market	0.0070*** (0.0013)			
Wholesale cost x HHI, std.		0.0021*** (0.0006)		
Wholesale cost x rival count, std.			-0.0026*** (0.0006)	
Wholesale cost x independent share, std.				0.0002 (0.0006)
N	604,770	604,770	604,770	570,430
Within R2	0.159	0.159	0.159	0.164
Outcome mean	2.886	2.886	2.886	2.887

Notes: Clustered standard errors by station are in parentheses. The dependent variable is retail price. All models include station and calendar-week fixed effects. *** p<0.01, ** p<0.05, * p<0.10.

The magnitude of these interactions is important to keep in perspective. Concentration does not completely transform the pricing rule followed by gasoline stations. Rather, local competitive structure remains conditionally associated with a small but detectable difference in pass-through. This is the narrower claim supported by the main results.

7. Robustness and Extensions

The first robustness check asks whether this pattern depends upon the particular circle drawn around a station. Table 4 therefore repeats the interaction designs with one-, three-, and five-mile markets. The high-HHI interactions are 0.0046, 0.0070, and 0.0108, respectively, and each is statistically significant at the 1 percent level. The standardized rival-count interactions are -0.0021, -0.0026, and -0.0039, and each is likewise significant at the 1 percent level. The independent-share interactions, however, are 0.0008, 0.0002, and 0.0004, none of which is statistically significant at conventional levels. Thus, the stable signs for HHI and rival count

make the result less dependent upon a single geographic boundary, while the independent-share evidence remains weak at every radius.

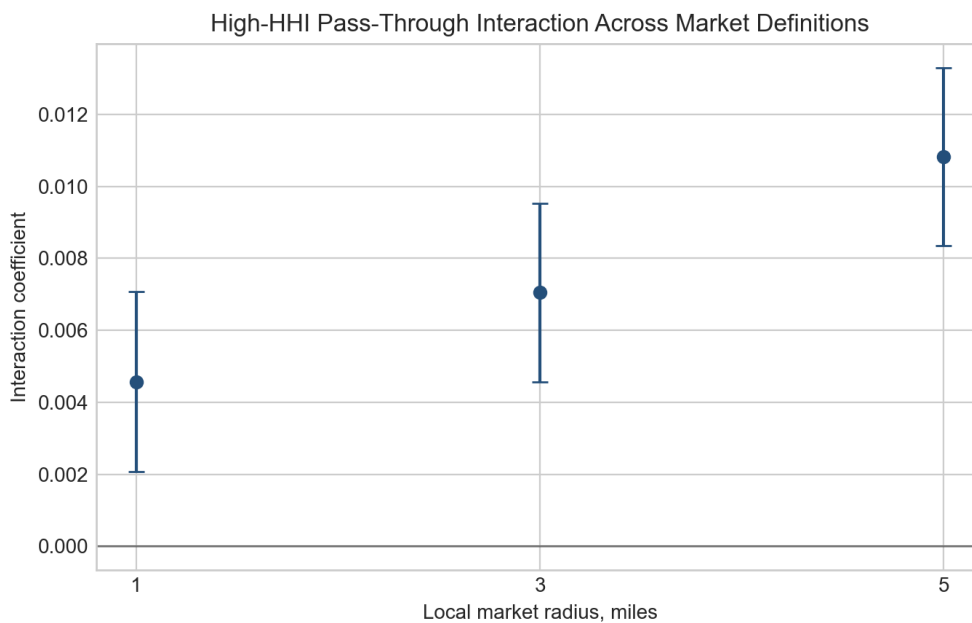
Table 4. Robustness to Alternative Geographic Market Definitions

Variable	High HHI 1 mi.	High HHI 3 mi.	High HHI 5 mi.	Rivals, std. 1 mi.	Rivals, std. 3 mi.	Rivals, std. 5 mi.	Ind. share, std. 1 mi.	Ind. share, std. 3 mi.	Ind. share, std. 5 mi.
Wholesale cost	0.5167*** (0.0047)	0.5157*** (0.0048)	0.5130*** (0.0049)	0.5193*** (0.0046)	0.5193*** (0.0046)	0.5191*** (0.0047)	0.5302*** (0.0049)	0.5264*** (0.0047)	0.5232*** (0.0046)
Cost x high HHI, 1 mi.	0.0046*** (0.0013)								
Cost x high HHI, 3 mi.		0.0070*** (0.0013)							
Cost x high HHI, 5 mi.			0.0108*** (0.0013)						
Cost x rivals, std., 1 mi.				-0.0021*** (0.0006)					
Cost x rivals, std., 3 mi.					-0.0026*** (0.0006)				
Cost x rivals, std., 5 mi.						-0.0039*** (0.0006)			
Cost x ind. share, std., 1 mi.							0.0008 (0.0007)		
Cost x ind. share, std., 3 mi.								0.0002 (0.0006)	
Cost x ind. share, std., 5 mi.									0.0004 (0.0006)
N	604,770	604,770	604,770	604,770	604,770	604,770	499,279	570,430	593,899
Within R2	0.159	0.159	0.160	0.159	0.159	0.159	0.169	0.164	0.161

Notes: Clustered standard errors by station are in parentheses. The dependent variable is retail price. All models include station and calendar-week fixed effects. *** p<0.01, ** p<0.05, * p<0.10. The radius and market-structure measure vary across columns.

Figure 2 presents the same high-HHI comparison visually. The point is not to claim that one of these radii is the true legal market, but to ask whether the estimate disappears when the circle around the station is drawn somewhat more narrowly or broadly. Changing the radius cannot replace a full SSNIP-style geographic-

market exercise. Nevertheless, given the data available here, it provides a useful test of whether the result is tied to one arbitrary definition.



It must also be remembered that these measures describe different objects. HHI measures brand concentration, rival count measures the number of nearby alternatives, and independent share measures the type of those alternatives. For this reason, they need not produce exactly the same coefficient. Reading them together gives a fuller picture than relying upon HHI alone.

Table 5 turns to the timing of pass-through. The current, first-lag, and second-lag wholesale-cost coefficients are 0.3402, 0.1267, and 0.1144, respectively, and all three are statistically significant at the 1 percent level. Their cumulative coefficient for below-median-HHI stations is therefore 0.5813. The current high-HHI interaction is -0.0188 and significant at the 1 percent level; the first-lag interaction is 0.0000 and not statistically significant at conventional levels; and the second-lag interaction is 0.0265 and significant at the 1 percent level. Together, the three interactions sum to approximately 0.0077, producing cumulative pass-through of about 0.5890 for high-HHI stations over the included weeks. High-HHI stations therefore show lower pass-through in the current week, no meaningful difference in the first lag, and a larger response in the second lag. Concentration appears to alter the timing of pass-through, rather than simply increasing it at every horizon.

Table 5. Dynamic Pass-Through by Local Concentration

Variable	Dynamic high HHI
Wholesale cost	0.3402*** (0.0068)
Wholesale cost, lag 1	0.1267*** (0.0057)
Wholesale cost, lag 2	0.1144*** (0.0049)
Wholesale cost x high-HHI market	-0.0188*** (0.0051)
Lag 1 wholesale cost x high-HHI market	0.0000 (0.0047)
Lag 2 wholesale cost x high-HHI market	0.0265*** (0.0055)
N	591,724
Within R2	0.182
Outcome mean	2.879

Notes: Clustered standard errors by station are in parentheses. The model includes current and lagged wholesale costs, station and calendar-week fixed effects. *** p<0.01, ** p<0.05, * p<0.10.

Finally, Figure 3 and Table 6 approach the question through retail margins. The local-HHI coefficient is 0.0765 with a standard error of 0.0087 and is statistically significant at the 1 percent level. Because HHI ranges from zero to one, a 0.10 increase is associated with about \$0.0077 per gallon in higher margins. The rival-count coefficient is 0.0003 and significant at the 1 percent level, while the independent-share coefficient is -0.0015 and not significant at conventional levels. The major-brand and independent-station coefficients are 0.0611 and 0.0241, and both are significant at the 1 percent level. A \$10,000 increase in tract income per capita is associated with a 0.0233 increase in margin and is significant at the 1 percent level, whereas the tract-unemployment coefficient is 0.0002 and not statistically significant. These coefficients do not all point toward one simple market-power story. For that reason, the margin model must remain a complementary piece of evidence.

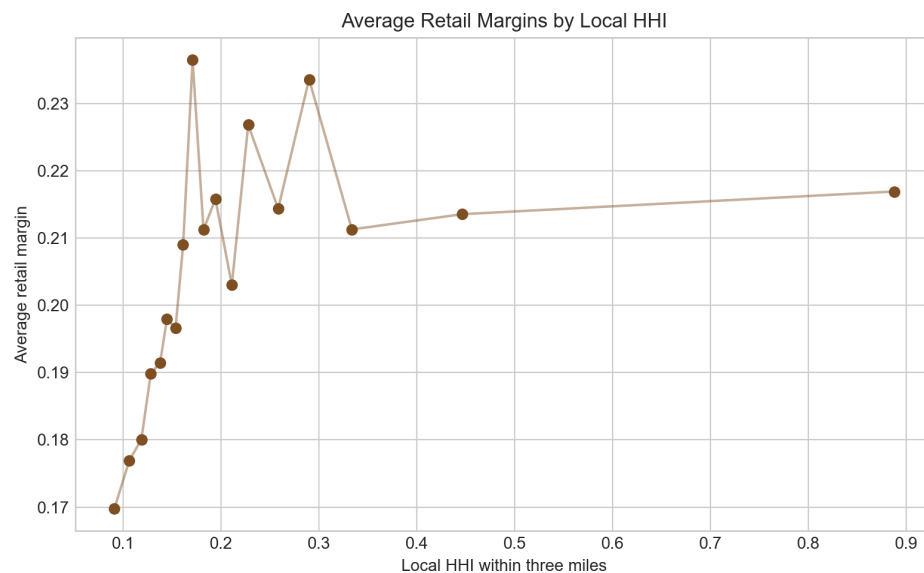


Table 6. Retail Margins and Local Market Structure

Variable	Margin model
Local HHI, 3 miles	0.0765*** (0.0087)
Nearby rival stations, 3 miles	0.0003*** (0.0001)
Nearby independent share, 3 miles	-0.0015 (0.0043)
Major-brand station	0.0611*** (0.0028)
Independent station	0.0241*** (0.0024)
Tract income per capita, \$10,000s	0.0233*** (0.0015)
Tract unemployment rate	0.0002 (0.0003)
N	568,346
Within R2	0.109
Outcome mean	0.205

Notes: Clustered standard errors by station are in parentheses. The dependent variable is the retail margin. The model includes calendar-week fixed effects and observed station and local controls. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The margin evidence is also less causally clean than the station fixed-effects evidence. Concentration is not randomly assigned, and a station in a concentrated market may differ from a station in an unconcentrated market in ways that the data do not capture. Entry itself may respond to expected margins, creating the possibility of reverse causality. Thus, the margin result is consistent with a market-power interpretation, but it cannot by itself establish that concentration caused the higher margin.

8. Antitrust Interpretation

Using the above results, we can now return to the antitrust question. If pass-through differs with concentration, then local competition affects not only the average price that consumers face, but also the timing and degree to which upstream cost shocks reach them. Gasoline shocks are both frequent and visible. In this way, the discretion that a station has when responding to those shocks becomes relevant to consumer welfare.

This evidence, of course, would not stand alone in an antitrust investigation. An agency or court would also want ownership data, quantities, evidence of consumer substitution and entry, and perhaps internal documents describing pricing decisions. A merger case would additionally require diversion evidence or another method of predicting where lost sales would go. Yet a systematic difference in pricing responses can still help reveal the competitive constraints that firms actually experience.

Thus, these estimates neither prove collusion nor simulate the effect of a merger. They also cannot fully separate the market-power hypothesis from a differential-efficiency explanation. The claim supported by the paper is narrower: station-level pricing behavior is systematically related to local competitive conditions. Although this would be only one part of a broader investigation, it remains meaningful evidence about market power in a geographically local retail market.

9. Conclusion

The limitations of the analysis must be understood before its conclusion can be stated. The data contain no quantities, market structure is observational, and the sample covers only Kentucky and Virginia. HHI is based on brand rather than ownership, while the geographic markets are approximations rather than formal SSNIP tests. The estimates must therefore be read as systematic conditional relationships, not as a complete welfare analysis. A stronger design could observe a change in market structure directly and incorporate ownership, traffic, or external measures of demand.

Nevertheless, this paper finds a modest relationship between local gasoline market structure and cost pass-through. In the station and week fixed-effects models, concentration and rival density appear within the pricing equation itself, rather than existing only as static descriptions of the market. By understanding market power as a question about the constraints firms face, the empirical results can be connected back to the

antitrust concern with which the paper began. Local structure does not determine conduct completely, but neither is it absent from that conduct. It leaves a small, measurable trace in how wholesale costs reach the prices paid by consumers.

Works Cited

- Ashenfelter, Orley, Daniel Hosken, and Matthew Weinberg. 2015. "Efficiencies Brewed: Pricing and Consolidation in the U.S. Beer Industry." *The RAND Journal of Economics* 46 (2): 328-361. <https://doi.org/10.1111/1756-2171.12092>.
- Baker, Jonathan B. 1999. "Econometric Analysis in FTC v. Staples." *Journal of Public Policy & Marketing* 18 (1): 11-21. <https://doi.org/10.1177/074391569901800103>.
- Borenstein, Severin, A. Colin Cameron, and Richard Gilbert. 1997. "Do Gasoline Prices Respond Asymmetrically to Crude Oil Price Changes?" *The Quarterly Journal of Economics* 112 (1): 305-339. <https://doi.org/10.1162/003355397555118>.
- Borenstein, Severin, and Andrea Shepard. 1996. "Dynamic Pricing in Retail Gasoline Markets." *The RAND Journal of Economics* 27 (3): 429-451. <https://www.jstor.org/stable/2555838>
- Dafny, Leemore, Mark Duggan, and Subramaniam Ramanarayanan. 2012. "Paying a Premium on Your Premium? Consolidation in the U.S. Health Insurance Industry." *American Economic Review* 102 (2): 1161-1185. <https://doi.org/10.1257/aer.102.2.1161>.
- U.S. Department of Justice and Federal Trade Commission. 2010. *Horizontal Merger Guidelines*. <https://www.justice.gov/atr/horizontal-merger-guidelines-08192010>.
- U.S. Department of Justice and Federal Trade Commission. 2023. *Merger Guidelines*. <https://www.justice.gov/atr/merger-guidelines>.
- Hastings, Justine S. 2004. "Vertical Relationships and Competition in Retail Gasoline Markets: Empirical Evidence from Contract Changes in Southern California." *American Economic Review* 94 (1): 317-328. <https://doi.org/10.1257/000282804322970823>.
- Houde, Jean-François. 2012. "Spatial Differentiation and Vertical Mergers in Retail Markets for Gasoline." *American Economic Review* 102 (5): 2147-2182. <https://doi.org/10.1257/aer.102.5.2147>.

- Kaplow, Louis, and Carl Shapiro. 2007. "Antitrust." In *Handbook of Law and Economics*, edited by A. Mitchell Polinsky and Steven Shavell, 1073-1225. Amsterdam: Elsevier. [https://doi.org/10.1016/S1574-0730\(07\)02015-4](https://doi.org/10.1016/S1574-0730(07)02015-4).
- Nakamura, Emi, and Dawit Zerom. 2010. "Accounting for Incomplete Pass-Through." *The Review of Economic Studies* 77 (3): 1192-1230. <https://doi.org/10.1111/j.1467-937X.2009.589.x>.
- Viscusi, W. Kip, Joseph E. Harrington, Jr., and David E. M. Sappington. 2018. *Economics of Regulation and Antitrust*. 5th ed. Cambridge, MA: MIT Press.
- Weyl, E. Glen, and Michal Fabinger. 2013. "Pass-Through as an Economic Tool: Principles of Incidence under Imperfect Competition." *Journal of Political Economy* 121 (3): 528-583. <https://doi.org/10.1086/670401>.
- Williamson, Oliver E. 1968. "Economies as an Antitrust Defense: The Welfare Tradeoffs." *American Economic Review* 58 (1): 18-36.